Majorana Modes in the Machine

Block 2: The Qubit Encoding

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Outline

Why map fermions to qubits?

Goal: simulate the Kitaev chain on a quantum computer.

Quantum computers operate on **qubits** (two-level systems), not directly on fermions.

Problem: fermionic operators anticommute, qubit operators on different sites commute.

$$\{c_i, c_j^\dagger\} = \delta_{ij} \quad \text{vs} \quad [\sigma_i^\alpha, \sigma_j^\beta] = 0 \; (i \neq j)$$

Jordan-Wigner solution

Attach a **string** of Z operators to each fermionic operator. The string encodes the fermion occupancy of all preceding sites and restores the correct anticommutation.

The Jordan-Wigner mapping

$$c_j = \underbrace{\left(\prod_{k=0}^{j-1} Z_k\right)}_{\text{JW string}} \sigma_j^-, \qquad c_j^\dagger = \left(\prod_{k=0}^{j-1} Z_k\right) \sigma_j^+ \tag{1}$$

where $\sigma^\pm = \frac{1}{2}(X \pm iY)$ are the qubit raising/lowering operators, X, Y, Z are the standard Pauli matrices, and Z_k acts only on qubit k.

Key identities

$$c_j^\dagger c_j = \frac{I - Z_j}{2} \quad \text{(number op.)}$$

$$c_j^\dagger c_{j+1} + \mathrm{h.c.} = -\frac{X_j X_{j+1} + Y_j Y_{j+1}}{2}$$

$$c_j^\dagger c_{j+1}^\dagger + \mathrm{h.c.} = -\frac{X_j X_{j+1} - Y_j Y_{j+1}}{2}$$

Why the string works

Z_k flips sign when site k is occupied. For $i < j$: $(c_i^\dagger)(c_j)$ picks up $Z_i \cdots Z_{j-1}$ from the JW string of c_j, turning commutation into *anti*commutation.

Substituting JW into H_K

Applying the identities to each term of
$H_K = -\mu \sum_j c_j^\dagger c_j - t \sum_j (c_j^\dagger c_{j+1} + \mathrm{h.c.}) + \Delta \sum_j (c_j^\dagger c_{j+1}^\dagger + \mathrm{h.c.})$:

$$\boxed{H_K = -\frac{\mu}{2} \sum_j Z_j + \frac{t-\Delta}{2} \sum_j X_j X_{j+1} + \frac{t+\Delta}{2} \sum_j Y_j Y_{j+1}} \qquad (2)$$

Structure

- $\mu \to$ **transverse field** (Z terms)
- Hopping $t \to XX + YY$ (number-conserving)
- Pairing $\Delta \to XX - YY$ (number-breaking)

Sweet spot $t = \Delta$

The XX terms cancel.

$$H = -\frac{\mu}{2} \sum_j Z_j + t \sum_j Y_j Y_{j+1}$$

A pure transverse-field + YY model.

Fermion parity is conserved

The **fermion parity operator**

$$P = \prod_{j=0}^{L-1} Z_j, \qquad P^2 = I, \quad [H_K, P] = 0 \tag{3}$$

splits the 2^L-dimensional Hilbert space into two sectors:

- **Even** parity ($P = +1$): even number of fermions
- **Odd** parity ($P = -1$): odd number of fermions

Topological degeneracy in the qubit language

In the topological phase the Majorana zero modes hybridize the two parity sectors. The lowest even- and odd-parity states become **degenerate** — the *parity gap* $|E_0^{(+)} - E_0^{(-)}| \to 0$.

This gap is exactly the Majorana hybridization splitting $E_0(L)$ from Block 1, now visible as a *spectral* signature in qubit space.

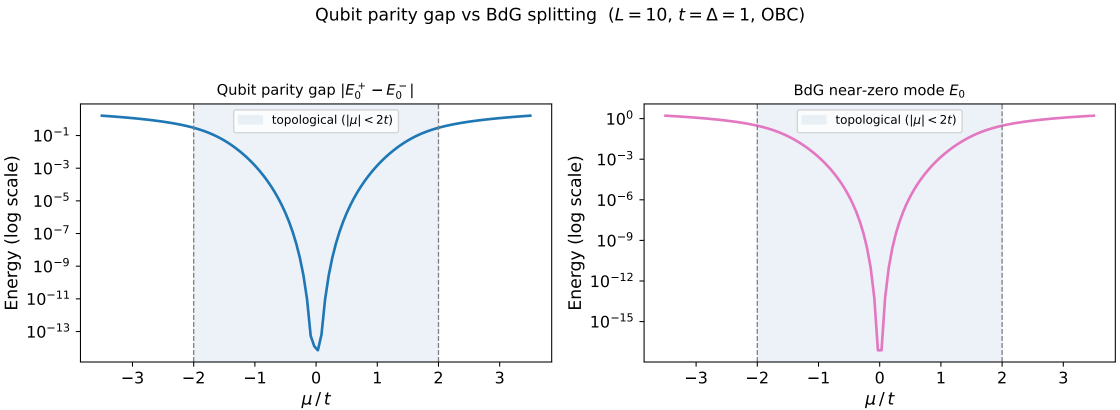
Building the qubit Hamiltonian in code

```
def kitaev_qubit_hamiltonian(L, t=1.0, mu=0.0, delta=1.0):
    dim = 2 ** L
    H = np.zeros((dim, dim), dtype=complex)

    for j in range(L):
        H -= (mu / 2) * pauli_at(Z, j, L)          # transverse field

    for j in range(L - 1):
        H += ((t - delta) / 2) * two_site(X, X, j, L)   # XX coupling
        H += ((t + delta) / 2) * two_site(Y, Y, j, L)   # YY coupling

    return H
```

Tensor-product helpers

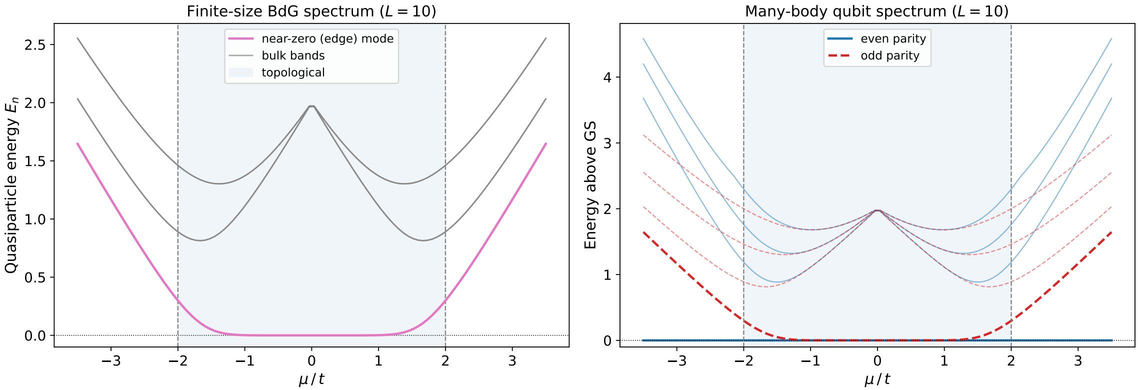
`pauli_at(op, j, L)` embeds a single-site Pauli at site j via $I \otimes \cdots \otimes \mathrm{op}_j \otimes \cdots \otimes I$.
`two_site(A, B, j, L)` places $A_j \otimes B_{j+1}$. Both are thin wrappers around `np.kron`.

Parity gap (qubit) vs BdG splitting — cross-check



Both methods give the same exponential decay in the topological phase and the same gap size in the trivial phase — confirming the JW mapping is exact.

Many-body qubit spectrum vs BdG spectrum



Even (blue) and odd (red) parity sectors are degenerate inside the topological phase and split outside — the bulk-boundary correspondence is manifest directly in the many-body spectrum alongside the BdG modes.

Block 2 — key takeaways

1. The **Jordan-Wigner transform** maps fermions to qubits by attaching a Z-string that restores anticommutation.
2. The Kitaev Hamiltonian becomes an **XY model with transverse field**: $-\frac{\mu}{2}\sum Z_j + \frac{t-\Delta}{2}\sum XX + \frac{t+\Delta}{2}\sum YY$.
3. **Fermion parity** $P = \prod Z_j$ is conserved — it partitions the spectrum into even/odd sectors.
4. The **parity gap** $|E_0^+ - E_0^-|$ is exponentially small in the topological phase and of order the bulk gap in the trivial phase — a direct qubit-space signature of topology.
5. Qubit and BdG spectra **agree exactly** — validating the mapping numerically.

Next (Block 3): Measuring topology — circuit-based measurements, edge & string observables.